

Lecture 15

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15.1 Poles & Zeros of MIMO Systems

15.1.1 Poles & Zeros of SISO Systems

Let $G(s)$ (or $G(z)$ in DT case) and $\left(\frac{A}{C} \middle| \frac{B}{D}\right)$ be the transfer function and a *minimal* state-space representation of a SISO LTI system.

p_0 is a pole of the system if

- $\lim_{s \rightarrow p_0} G(s) = \infty$
- p_0 is an eigenvalue of A

whereas z_0 is a zero of the system if

- $\lim_{s \rightarrow z_0} G(s) = 0$
- $\exists x_0$ such that output response to $u(t) = e^{z_0 t}$ and $x(0) = x_0$ satisfies $y(t) = 0$

Let's remember the structure of the response for input types in the form of $u(t) = e^{z_0 t}$

$$y(t) = \underbrace{C e^{At} [\bar{x}_0 + x_0]}_{\text{transient}} + \underbrace{G(z_0) e^{z_0 t}}_{\text{steady-state}}, \text{ where}$$

$$\bar{x}_0 = [-(z_0 I - A)^{-1} B] \quad , \quad G(z_0) = C(z_0 I - A)^{-1} B + D$$

Thus z_0 is a zero if

$$x_0 = -\bar{x}_0 = (z_0 I - A)^{-1} B$$

$$G(z_0) = C(z_0 I - A)^{-1} B + D = 0$$

where x_0 is called the **state-zero-direction** associated with z_0 .

Note that these definitions and conditions directly hold for DT SISO LTI systems.

15.1.2 Poles of MIMO Systems

Unlike MIMO zeros, the definition and derivation of MIMO poles are much more straightforward.

Let $G(s)$ (or $G(z)$ in DT case) and $\left(\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right)$ be the transfer function matrix and a *minimal* state-space representation of a MIMO LTI system.

p_0 is a pole of the system if

- $\exists(i, j)$ s.t. $\lim_{s \rightarrow p_0} G_{ij}(s) = \infty$
- $\lim_{s \rightarrow p_0} \|G(s)\| = \infty$
- p_0 is an eigenvalue of A

In other words, in the context of the transfer function matrix, p_0 is a pole of the system if it is a pole of any entry of $G(s)$. Understanding and deriving the multiplicities of a pole based on the transfer function matrix is a little bit tricky and not very intuitive. Rigorously, this is done using the **Smith-McMillan Form** of the transfer matrix. However, for the context and scope of this class, we can state that we can find a pole's algebraic and geometric multiplicity based on the Jordan decomposition of A , provided that the state-space representation is minimal.

Ex 15.1

$$G(s) = \begin{bmatrix} \frac{s+1}{(s+2)^2} & 0 & 0 \\ 0 & \frac{s}{(s+1)(s+2)} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We can see that $p_1 = -1$ and $p_2 = -2$ are the system's poles. Note that $z = -1$ is also a zero of the first entry of the transfer function matrix (and indeed, it is a zero of the system). This states that a MIMO system can have a pole and a zero at the exact same location. ■

15.1.3 Zeros of MIMO Systems

Suppose we generalize the definition of a zero in SISO systems to MIMO systems using the same logic as MIMO poles. In that case, we can see that the resultant definition would not be helpful and somewhat unsatisfactory. Let's consider the transfer function matrix in the previous example. In this example, all non-diagonal elements are zero, and hence if we define a zero such that $\exists(i, j)$ s.t. $\lim_{s \rightarrow z_0} G_{ij}(s) = 0$, then any $z_0 \in \mathbb{C}$ would be a zero of the system. In this context, we need a more useful and satisfactory definition(s) for MIMO zeros.

Definition: Let $z_0 \neq p_i, \forall p_i \in \mathcal{P}$, where $\mathcal{P}$ is the set of all poles of the MIMO system $G(s)$ with m inputs and q outputs. z_0 is a zero of the system (often called a **Transmission Zero**) if $G(s)$ drops rank at $s = z_0$.

Naturally, assuming that $G(s)$ has full **normal rank** (i.e., it is full rank for *most* $s \in \mathbb{C}$), we can also observe that:

- if $m \leq q$, $G(s)$ drops rank at $s = z_0 \iff \exists u_0 \in \mathbb{R}^m$ & $u_0 \neq 0$ such that $G(z_0)u_0 = 0$ (**Right Zeros**)
- if $q \leq m$, $G(s)$ drops rank at $s = z_0 \iff \exists w_0^T \in \mathbb{R}^q$ & $w_0 \neq 0$ such that $w_0^T G(z_0) = 0$ (**Left Zeros**)

These definitions provide us the zeros different from the system's poles. u_0 and w_0 are called the **input-zero-directions** of a MIMO zero in their respective cases.

Note on system shape: Systems with more inputs than outputs ($m > q$) have redundant control authority. This redundancy physically allows us to "hide" certain dynamics from the output by carefully coordinating the inputs along specific zero directions.

Ex 15.2

$$G(s) = \begin{bmatrix} \frac{s+1}{(s+2)^2} & \frac{s+3}{s+4} \\ 0 & \frac{s}{s+2} \end{bmatrix}$$

Solution: In this example, we can see that $G(s)$ is full normal rank for *most* $s \in \mathbb{C}$. The matrix drops rank at locations $z_0 \in \{0, -1\}$. However, the rank is preserved for -3 even if $G_{12}(-3) = 0$. Thus, the zeros of the system are $\{0, -1\}$. ■

Let's solve some non-square examples.

Ex 15.3

$$G(s) = \begin{bmatrix} \frac{s+1}{(s+2)^2} & \frac{s+3}{s+4} \\ 0 & \frac{s}{s+2} \\ \frac{s(s+3)}{(s+4)^2} & 0 \end{bmatrix}$$

Solution: Let's test if $s = -1$ is a zero, and if it is, let's find u_{-1} such that $G(-1)u_{-1} = 0$:

$$G(-1) = \begin{bmatrix} 0 & \frac{2}{3} \\ 0 & -1 \\ -\frac{2}{9} & 0 \end{bmatrix} \Rightarrow \text{rank}[G(-1)] = 2$$

Thus, for this system, $s = -1$ is not a zero. Let's test if $s = -3$ is a zero:

$$G(-3) = \begin{bmatrix} -2 & 0 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{rank}[G(-3)] = 2$$

Thus, also for this system, $s = -3$ is not a zero. Let's test if $s = 0$ is a zero, and if it is, let's find u_0 such that $G(0)u_0 = 0$:

$$G(0) = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{rank}[G(0)] = 1$$

To find the null space, we solve $G(0)u_0 = 0$, which yields the equation $\frac{1}{4}u_1 + \frac{3}{4}u_2 = 0 \Rightarrow u_1 = -3u_2$. Choosing $u_2 = 4$, we get $u_1 = -12$.

$$u_0 = \begin{bmatrix} -12 \\ 4 \end{bmatrix} \Rightarrow G(0)u_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus $s = 0$ is a zero of the system and the **direction** of the zero is $u_0 = \begin{bmatrix} -12 \\ 4 \end{bmatrix}$. ■

Ex 15.4

$$G(s) = \begin{bmatrix} \frac{s+1}{(s+2)^2} & \frac{s+3}{s+4} & 0 \\ 0 & \frac{s}{s+2} & \frac{(s+1)(s+3)}{(s-1)^2} \end{bmatrix}$$

Solution: Let's test if $s = -1$ is a zero, and if it is, let's find w_{-1} such that $w_{-1}^T G(-1) = 0$:

$$G(-1) = \begin{bmatrix} 0 & \frac{2}{3} & 0 \\ 0 & -1 & 0 \end{bmatrix} \Rightarrow \text{rank}[G(-1)] = 1$$

We look for $w_{-1} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such that $\frac{2}{3}w_1 - 1w_2 = 0 \Rightarrow 2w_1 = 3w_2$.

$$w_{-1} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \Rightarrow w_{-1}^T G(-1) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Thus $s = -1$ is a left zero of the system and the **direction** of the zero is $w_{-1} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Now let's test if $s = -3$ and $s = 0$ are zeros of the system:

$$G(-3) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \Rightarrow \text{rank}[G(-3)] = 2$$

$$G(0) = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \text{rank}[G(0)] = 2$$

Thus neither $s = -3$ nor $s = 0$ is a zero of the system. ■

The major limitation of this definition is that it is limited to zeros that are different from the poles of the system. So we need a more general definition of zeros.

Definition: Let's assume that $G(s)$ has full normal rank with m inputs and q outputs. $z_0 \in \mathbb{C}$ is a zero of the system if:

- if $m \leq q$, $\exists u(s) \neq 0$ such that $\lim_{s \rightarrow z_0} [G(s)u(s)] = 0$ (**Right Zeros**)
- if $q \leq m$, $\exists w(s) \neq 0$ such that $\lim_{s \rightarrow z_0} [w(s)^T G(s)] = 0$ (**Left Zeros**)

Ex 15.5

$$G(s) = \begin{bmatrix} 1 & \frac{s+1}{s+2} \\ 0 & 1 \end{bmatrix}$$

Let $u(s) = \begin{bmatrix} 1 \\ s+2 \end{bmatrix}$, then

$$\lim_{s \rightarrow -2} [G(s)u(s)] = \lim_{s \rightarrow -2} \begin{bmatrix} s+2 \\ s+2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus $s = -2$ is both a pole and a zero location for $G(s)$. Physically, this means the zero direction perfectly aligns in a way that blocks the excitation of that specific pole from reaching the output.

If you are bothered by the fact that the selected input $u(s)$ is non-causal, you can simply multiply $u(s)$ with $\frac{1}{s+b}$ where b is not a zero or pole of the system, creating a causal input. We will still observe the cancellation:

$$\bar{u}(s) = \frac{u(s)}{s+b} = \left[\frac{\frac{1}{s+b}}{\frac{s+2}{s+b}} \right]$$

$$\lim_{s \rightarrow -2} [G(s)\bar{u}(s)] = \lim_{s \rightarrow -2} \left[\frac{\frac{s+2}{s+b}}{\frac{s+2}{s+b}} \right] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

■

Corollary: Let $G(s)$ be a square transfer function matrix and it is invertible for *most* $s \in \mathbb{C}$, then the zeros of the system are exactly the poles of $G(s)^{-1}$.

Let's analyze the previous example:

$$G(s)^{-1} = \begin{bmatrix} 1 & -\frac{s+1}{s+2} \\ 0 & 1 \end{bmatrix}$$

$$\lim_{s \rightarrow -2} \|G(s)^{-1}\| = \infty \Rightarrow z_0 = -2 \text{ is a zero of } G(s)$$

15.1.3.1 Determining Zeros from a Minimal Realization

Physical Interpretation of a Zero Direction: Consider a system with two actuators (inputs) and one sensor (output) controlling a structural vibration. A right zero at $s = j\omega$ with direction u_0 means that if we actuate the system at frequency ω using the specific proportional combination of actuators defined by u_0 , the internal forces will perfectly cancel out exactly at the sensor location, yielding zero steady-state output.

Given a minimal state realization of a MIMO system with quadruple (A, B, C, D) , z_0 is an **Invariant Zero** of the system if it drops the rank of the **Rosenbrock System Matrix**, given below:

$$\mathbf{S}(s) = \left[\begin{array}{c|c} sI - A & -B \\ \hline C & D \end{array} \right]$$

Notice that for an observable and reachable system, $\mathbf{S}(s)$ is full rank for *most* $s \in \mathbb{C}$. Let's first verify that this condition correctly gives the zeros of a SISO system. Let's assume that $\mathbf{S}(s)$ drops rank at z_0 , then we know that $\exists v_0 = \begin{bmatrix} x_0 \\ u_0 \end{bmatrix}$, where $x_0 \in \mathbb{R}^n$ and $u_0 \in \mathbb{R}$ such that

$$\mathbf{S}(z_0)v_0 = 0 \Rightarrow \left[\begin{array}{c|c} z_0I - A & -B \\ \hline C & D \end{array} \right] \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0 \Rightarrow \begin{cases} [z_0I - A]x_0 - Bu_0 = 0 \\ Cx_0 + Du_0 = 0 \end{cases}$$

Note that for SISO systems, zeros cannot be located at the pole locations, hence $[z_0I - A]x_0 \neq 0$, which implies that $u_0 \neq 0$. Let $u_0 = 1$, then

$$x_0 = [z_0I - A]^{-1} B$$

$$C[z_0I - A]^{-1} B + D = G(z_0) = 0$$

Note that these are minimal state-space representation-based conditions that provide the zeros of a SISO system. In other words, if $x(0) = x_0 = [z_0I - A]^{-1} B$ and $u(t) = e^{z_0 t}$, then $y(t) = 0$, $\forall t \geq 0$.

Now, let's verify that this condition correctly gives the zeros of a MIMO system that are different from the poles.

Let's first focus on the case $m \leq q$, where the number of inputs is less than or equal to the number of outputs. Let's assume that $\mathbf{S}(s)$ drops rank at z_0 , then we know that $\exists v_0 = \begin{bmatrix} x_0 \\ u_0 \end{bmatrix}$, where $x_0 \in \mathbb{R}^n$ and $u_0 \in \mathbb{R}^m$ such that

$$\begin{aligned} \mathbf{S}(z_0)v_0 = 0 &\Rightarrow \left[\begin{array}{c|c} z_0I - A & -B \\ \hline C & D \end{array} \right] \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0 \Rightarrow \begin{cases} [z_0I - A]x_0 - Bu_0 = 0 \\ Cx_0 + Du_0 = 0 \end{cases} \\ x_0 &= [z_0I - A]^{-1}Bu_0 \\ \left[C[z_0I - A]^{-1}B + D \right]u_0 &= G(z_0)u_0 = 0 \end{aligned}$$

Note that for a MIMO system with $m \leq q$, z_0 is a zero of the system that is different from the poles, if $\exists u_0 \in \mathbb{R}^m$ such that $G(z_0)u_0 = 0$, which is the condition that we derived earlier. Moreover, if we assume $x(0) = x_0$ and $u(t) = u_0 e^{z_0 t}$ we will see that $y(t) = 0 \forall t \geq 0$. x_0 and u_0 are called the **state-zero-direction** and **input-zero-direction** respectively associated with **right-zero** z_0 . This is indeed another more general definition of a MIMO zero (which also correctly holds for zeros located at the system's poles).

Let's focus on the case $q \leq m$, where the number of outputs is less than or equal to the number of inputs. Let's assume that $\mathbf{S}(s)$ drops rank at z_0 , then we know that $\exists v_0^T = [x_0^T \ w_0^T]$, where $x_0 \in \mathbb{R}^n$ and $w_0 \in \mathbb{R}^q$ such that

$$\begin{aligned} v_0^T \mathbf{S}(z_0) = 0 &\Rightarrow \begin{bmatrix} x_0^T & w_0^T \end{bmatrix} \left[\begin{array}{c|c} z_0I - A & -B \\ \hline C & D \end{array} \right] = 0 \Rightarrow \begin{cases} x_0^T [z_0I - A] + w_0^T C = 0 \\ x_0^T B + w_0^T D = 0 \end{cases} \\ [z_0I - A]^T x_0 + C^T w_0 = 0 &\Rightarrow x_0 = - \left([z_0I - A]^T \right)^{-1} C^T w_0 \\ x_0^T B + w_0^T D = 0 &\Rightarrow B^T x_0 + D^T w_0 = 0 \Rightarrow \left[-B^T \left([z_0I - A]^T \right)^{-1} C^T + D^T \right] w_0 = 0 \\ G(z_0)^T w_0 = 0 &\iff w_0^T G(z_0) = 0 \end{aligned}$$

Note that for a MIMO system with $q \leq m$, z_0 is a zero of the system that is different from the poles, if $\exists w_0 \in \mathbb{R}^q$ such that $w_0^T G(z_0) = 0$. Moreover, if we assume $x(0) = x_0$ and $w(t) = w_0 e^{z_0 t}$ for a dual system where the state-space matrices have the form (A^T, C^T, B^T, D^T) , we will see that $y(t) = 0 \forall t \geq 0$. x_0 and w_0 are called the **state-zero-direction** and **output-zero-direction** respectively associated with **left-zero** z_0 .

Ex 15.6

$$G(s) = \begin{bmatrix} 1 & \frac{s+1}{s+2} \\ 0 & 1 \end{bmatrix}$$

Previously we found that $G(s)$ has a pole and a zero at $s = -2$. Let's verify this using the Rosenbrock Matrix condition. We can find a minimal realization using Gilbert's method:

$$\begin{aligned} G(s) &= \begin{bmatrix} 1 & 1 - \frac{1}{s+2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \frac{1}{s+2} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \frac{1}{s+2} \\ A &= -2, B = \begin{bmatrix} 0 & 1 \end{bmatrix}, C = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, D = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Let's construct the Rosenbrock system matrix:

$$\mathbf{S}(s) = \left[\begin{array}{c|c} sI - A & -B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|cc} s+2 & 0 & -1 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right] \Rightarrow \text{rank}[\mathbf{S}(-2)] = 2$$

which verifies that the system has a single invariant zero at $s = -2$. Now let's find the direction of the zero.

$$\mathbf{S}(-2) \begin{bmatrix} x_0 \\ u_{0,1} \\ u_{0,2} \end{bmatrix} = 0$$

$$\left[\begin{array}{c|cc} 0 & 0 & -1 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right] \begin{bmatrix} x_0 \\ u_{0,1} \\ u_{0,2} \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} x_0 \\ u_{0,1} \\ u_{0,2} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Thus, $u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the input zero direction. ■